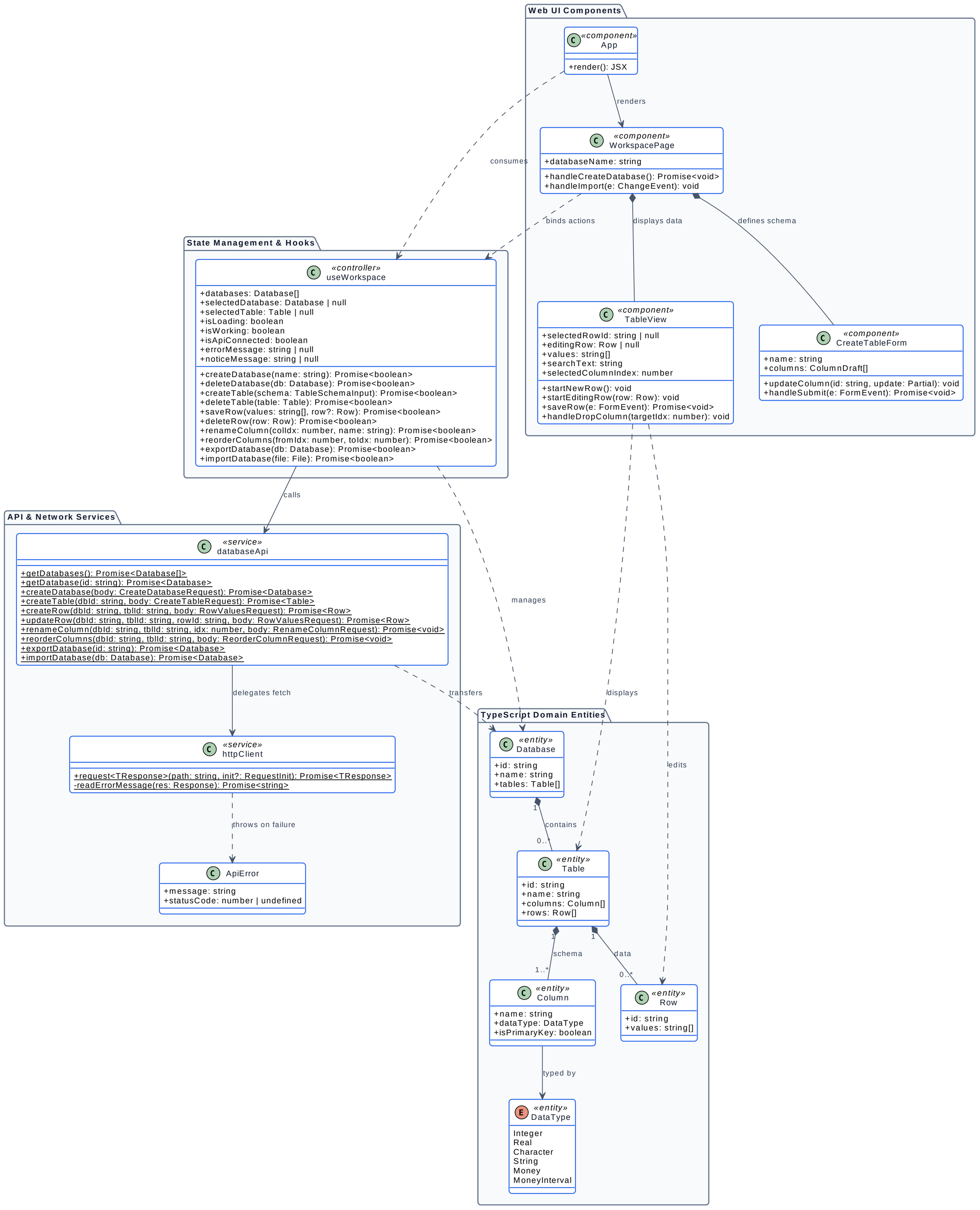
Web UI Components
«component»
App
+render(): JSX
renders
consumes
«component»
WorkspacePage
+databaseName: string
+handleCreateDatabase(): Promise<void>
+handleImport(e: ChangeEvent): void
binds actions
displays data
defines schema
«component»
TableView
+selectedRowId: string | null
+editingRow: Row | null
+values: string[]
+searchText: string
+selectedColumnIndex: number
+startNewRow(): void
+startEditingRow(row: Row): void
+saveRow(e: FormEvent): Promise<void>
+handleDropColumn(targetIdx: number): void
«component»
CreateTableForm
+name: string
+columns: ColumnDraft[]
+updateColumn(id: string, update: Partial): void
+handleSubmit(e: FormEvent): Promise<void>
State Management & Hooks
«controller»
useWorkspace
+databases: Database[]
+selectedDatabase: Database | null
+selectedTable: Table | null
+isLoading: boolean
+isWorking: boolean
+isApiConnected: boolean
+errorMessage: string | null
+noticeMessage: string | null
+createDatabase(name: string): Promise<boolean>
+deleteDatabase(db: Database): Promise<boolean>
+createTable(schema: TableSchemaInput): Promise<boolean>
+deleteTable(table: Table): Promise<boolean>
+saveRow(values: string[], row?: Row): Promise<boolean>
+deleteRow(row: Row): Promise<boolean>
+renameColumn(colIdx: number, name: string): Promise<boolean>
+reorderColumns(fromIdx: number, toIdx: number): Promise<boolean>
+exportDatabase(db: Database): Promise<boolean>
+importDatabase(file: File): Promise<boolean>
calls
manages
API & Network Services
«service»
databaseApi
+getDatabases(): Promise<Database[]>
+getDatabase(id: string): Promise<Database>
+createDatabase(body: CreateDatabaseRequest): Promise<Database>
+createTable(dbId: string, body: CreateTableRequest): Promise<Table>
+createRow(dbId: string, tblId: string, body: RowValuesRequest): Promise<Row>
+updateRow(dbId: string, tblId: string, rowId: string, body: RowValuesRequest): Promise<Row>
+renameColumn(dbId: string, tblId: string, idx: number, body: RenameColumnRequest): Promise<void>
+reorderColumns(dbId: string, tblId: string, body: ReorderColumnRequest): Promise<void>
+exportDatabase(id: string): Promise<Database>
+importDatabase(db: Database): Promise<Database>
delegates fetch
transfers
displays
edits
«service»
httpClient
+request<TResponse>(path: string, init?: RequestInit): Promise<TResponse>
-readErrorMessage(res: Response): Promise<string>
throws on failure
ApiError
+message: string
+statusCode: number | undefined
TypeScript Domain Entities
«entity»
Database
+id: string
+name: string
+tables: Table[]
1
contains
0..*
«entity»
Table
+id: string
+name: string
+columns: Column[]
+rows: Row[]
1
schema
1..*
1
data
0..*
«entity»
Column
+name: string
+dataType: DataType
+isPrimaryKey: boolean
«entity»
Row
+id: string
+values: string[]
typed by
«entity»
DataType
Integer
Real
Character
String
Money
MoneyInterval